

Instructor: Dr. Rajesh Kumble Nayak

01 Sep, 2025 @ 11:00 AM.

Duration 50 Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!

Problem - 1 {15 Marks}:In a generalised coordinate system r , the Lagrangian is given by

$$L(r, \dot{r}) = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2) - \frac{1}{2} \frac{mk}{a}(r - a)^2.$$

Here,

$$T = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2)$$

$$V = \frac{1}{2} \frac{mk}{a}(r - a)^2.$$

- Find the Lagrange equation of motion.
- Find at least one non-trivial conserved quantity.
- Is total energy E in this system.
- Find the solution to the equation of motion as an explicit function of time.

Problem - 2 {15 Marks}:Consider a Lagrangian, in a generalised coordinate system $\{\theta, \phi, \psi\}$, which is given by,

$$L = \frac{1}{2}I_1(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2}I_3(\dot{\psi} + \dot{\phi} \cos \theta)^2 - mgl \cos \theta,$$

A particle of mass m is confined to move along a vertically oriented circle under the influence of gravity. Find a generalised coordinate for this problem and write the Lagrangian for it.

- Find the constants of motion.
- Find the equation of motion.
- Is the system integrable?

Problem - 3 {15 Marks}:

Consider a free particle in a two-dimensional inertial frame,

- (1) Write the equation of motion in a plane polar coordinates.
- (2) Show that the particle moves along a straight line.

Instructor: Dr. Rajesh Kumble Nayak

31 Oct, 2025 @ 10:00 AM.

Duration 50 Min.

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- Keep answers concise and to the point.

Take a deep breath and relax.

Good luck!

Problem - 1 {15 Marks}:

Start from the Hamiltonian

$$H = \frac{1}{2m}p^2 + V(q),$$

with $m > 0$ and arbitrary smooth potential $V(q)$.

1. Show that the type-1 generating function $F_1(q, Q) = -qQ$, generates the canonical transformation that swaps coordinates and momenta, $Q = p$, and $P = q$. Write the transformation equations and verify them.
2. Write the transformed Hamiltonian $K(Q, P)$. Write Hamilton's equations of motion for Q, P and compare their form to the original equations for q, p .

Problem - 2 {15 Marks}:Let $U(q_i, p_i, t)$ be a smooth function in canonical variables, $\{q_i, p_i, H\}$ and time, find the Poisson bracket $[U, H]_{q,p}$.**Problem - 3 {15 Marks}:**A particle of mass m and charge e moves in the xy -plane in a uniform magnetic field $\mathbf{B} = B\hat{z}$. The Hamiltonian is

$$H(x, y, p_x, p_y) = \frac{1}{2m} \left[\left(p_x + \frac{eB}{2}y \right)^2 + \left(p_y - \frac{eB}{2}x \right)^2 \right].$$

The generalised momentum components are $\Pi_x := p_x + \frac{eB}{2}y$, $\Pi_y := p_y - \frac{eB}{2}x$, and the guiding-center coordinates is given by $X := x + \frac{\Pi_y}{eB}$, $Y := y - \frac{\Pi_x}{eB}$. Using $\{x, y, p_x, p_y\}$ as canonical variables,

1. Compute the Poisson brackets $[\Pi_x, \Pi_y]$.
2. Compute $[X, Y]$. Show the guiding-center coordinates do not Poisson-commute.
3. Compute $[X, H]$ and $[Y, H]$. Conclude whether (X, Y) are conserved as phase-space functions.

Instructor: Dr. Rajesh Kumble Nayak
19th, Nov 2025 @ 6:00 PM.
Duration 40 Min.

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- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Take a deep breath and relax.
Good luck!

Problem - 1 {20 Marks}:

For a Lagrangian given below:

$$\mathcal{L} = \frac{1}{2}ml^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2) - \frac{1}{2}mgl (\theta_1^2 + \theta_2^2) - \frac{1}{8}kl^2 (\theta_2 - \theta_1)^2.$$

1. Find the normal modes around the stable points.
2. Find the amplitudes corresponding to normal modes.

Problem - 2 {20 Marks}:

A physical system is described by the following kinetic and potential energy,

$$T = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2(\dot{x} + l\dot{\theta})^2 = \frac{1}{2}(m_1 + m_2)\dot{x}^2 + m_2l\dot{x}\dot{\theta} + \frac{1}{2}m_2l^2\dot{\theta}^2.$$

$$V = \frac{1}{2}k_1x^2 + \frac{1}{2}m_2gl\theta^2.$$

1. Find the normal modes around the stable points.
2. Find the amplitudes corresponding to normal modes.